M4: Final Deliverable

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***Abstract*—This project analyzes 32 million MovieLens ratings to answer these questions. The findings reveal that movies form a steep hierarchy: 94% are rarely watched, 0.8% are blockbuster hits, and just 21 films (0.07%) achieve true classic status. More surprisingly, what separates these tiers is not quality but popularity — a movie's rating count is 18 times more important than its average rating in determining its tier. For users, two distinct archetypes emerged: Engaged Raters (65%) who rate many movies but are critical, and Generous Raters (35%) who rate fewer movies but give higher ratings. What matters most is not how much users rate, but how they rate — rating quality and consistency are the strongest differentiators. Temporal analysis shows that all users rate more generously in their first session, then become more critical over time. These findings challenge assumptions about quality driving success and reveal that user engagement is multi-dimensional.**

I. Introduction

This project analyzes the MovieLens 32M dataset to uncover meaningful patterns in both movies and user behavior. The goal is to move beyond simple statistics and identify patterns in how movies group together, how users behave, and how engagement evolves over time. The findings have practical implications for content acquisition strategies, user engagement optimization, and recommendation system design.

A. Discovery Questions

- Q1: What natural groups of movies exist based on user ratings, genres, and tags?
- Q2: How do user engagement patterns (rating volume and tagging) relate to rating behavior, and how do these factors form distinct user archetypes?
- Q3: How do temporal patterns in rating activity (burst behavior, rating recency, and consistency over time) differ between user archetypes, and what anomalous rating patterns emerge?

B. Dataset

The MovieLens dataset contains approximately 32 million ratings, 2,000,072 tag applications, and 87,000 movies. For this analysis, data was aggregated at the movie level (average rating, rating count, genre indicators) and user level (rating behavior, tagging activity, temporal features). The dataset is anonymized with no personally identifiable information.

II. Methodology

A. Exploratory Data Analysis

Before preprocessing, exploratory data analysis was conducted to understand data quality and identify potential issues. Univariate analysis revealed that raw user ratings range from 0.5 to 5.0 in 0.5 increments, with a strong positive bias toward ratings of 4.0, 3.0, and 5.0. Half-star increments are used less frequently than whole stars, suggesting users tend to think in whole numbers. Genre analysis showed that Drama and Comedy represent approximately 40% of the library, while a small number of movies have "no genres listed," a category requiring handling during preprocessing.

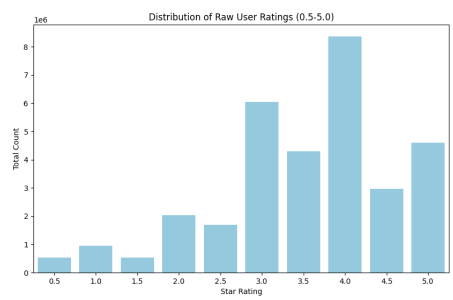


Fig. 1. Ratings Counts

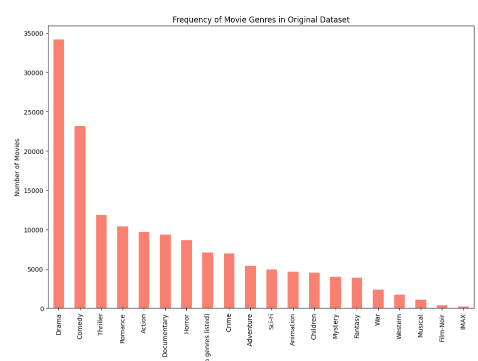


Fig. 2. Genres

Bivariate analysis revealed that median ratings remain remarkably stable across genres, indicating that genre alone is not a strong predictor of movie quality and justifying the use of additional dimensions such as popularity and tags in clustering.

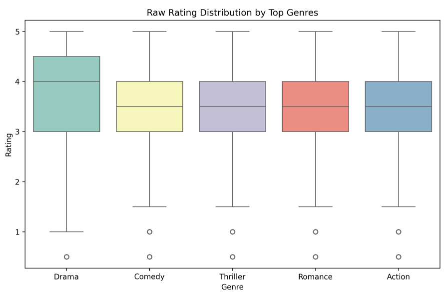


Fig. 3. Genre Distribution

B. Data Preprocessing

Movie features included average rating, rating count, popularity log (log-transformed rating count), release year, and 20 one-hot encoded genre indicators. User features included average rating, rating standard deviation, total ratings given, tags given, and maximum daily ratings. All features were standardized to have zero mean and unit variance prior to clustering. The preprocessing pipeline handles missing values, outliers (log-transformation for skewed distributions), and engineers behavioral features such as binge behavior and metadata richness.

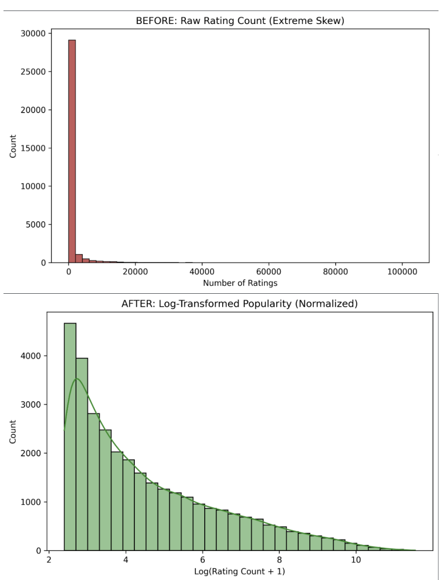


Fig. 4. Log. Transformation - Before vs. After

C. Clustering

Movies: K-Means clustering with PCA preprocessing identified four distinct clusters. The optimal number of clusters (k=4) was determined through silhouette score analysis (0.879) and the elbow method applied to the within-cluster sum of squares. PCA reduced the 24-dimensional feature space (average rating, popularity log, 20 genres) to 10 principal components that captured 90.2% of the total variance.

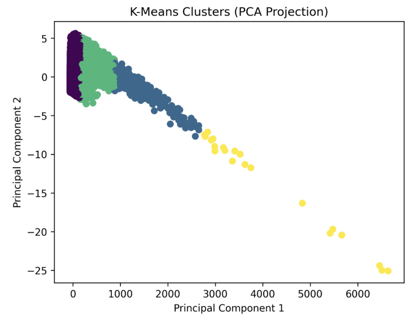


Fig. 5. Movie Clusters

Users: K-Means clustering on behavioral features identified two distinct archetypes. Silhouette analysis across k=2 through k=5 confirmed k=2 as optimal, with a silhouette score of 0.127 indicating modest but meaningful separation between groups. PCA reduced the feature space to 2 components for visualization, capturing 33.8% of the variance (17.5% from PC1, 16.4% from PC2).

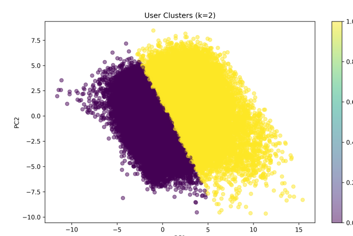
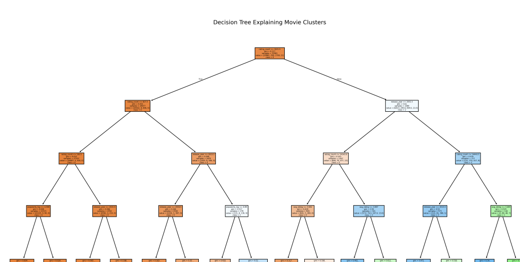


Fig. 6. User Clusters

D. Classification Validation

Decision trees were built using cluster assignments as class labels to validate that discovered groups represent real patterns rather than algorithmic artifacts:

Movies: The decision tree excluded the metadata_richness feature to avoid circular logic, using only average rating, popularity log, and genre indicators as predictors.

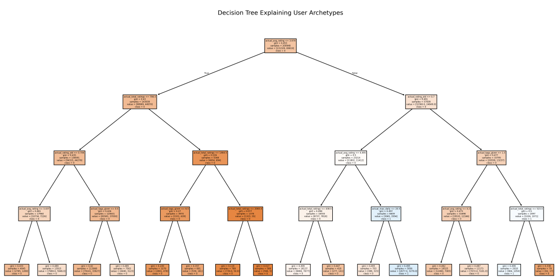


```
=== MOVIE CLUSTER FEATURE IMPORTANCE ===
rating_count            0.808883
release_year            0.140166
avg_rating              0.044212
popularity_log          0.006739
(no genres listed)      0.000000
Horror                  0.000000
Western                 0.000000
War                     0.000000
Thriller                0.000000
Sci-Fi                  0.000000
dtype: float64

=== MODEL ACCURACY ===
Decision Tree Accuracy: 0.956
Without metadata_richness: 0.956
```

Fig. 7. Movie Decision Tree

Users: The decision tree used only behavioral features (average rating, rating standard deviation, total ratings, tags given, maximum daily ratings), excluding genre preferences to directly address the discovery question about engagement patterns.




```
=== USER ARCHETYPE FEATURE IMPORTANCE ===
actual_avg_rating        0.516871
actual_rating_std        0.214588
actual_total_ratings     0.166317
actual_tags_given        0.097148
actual_max_daily         0.005076
dtype: float64
```

Fig. 8. User Decision Tree

The tree depth was limited to 4 levels (for both movies and users validation) to prioritize interpretability over predictive accuracy.

E. Temporal Analysis

Sessions were defined using a 2-hour gap threshold, selected after exploring 1-hour, 2-hour, and 4-hour options as a reasonable boundary between distinct rating sessions while being robust to brief interruptions. Analysis included first versus later session comparison, rating recency (years after movie release), user lifespan (days between first and last rating), and rating evolution over time.

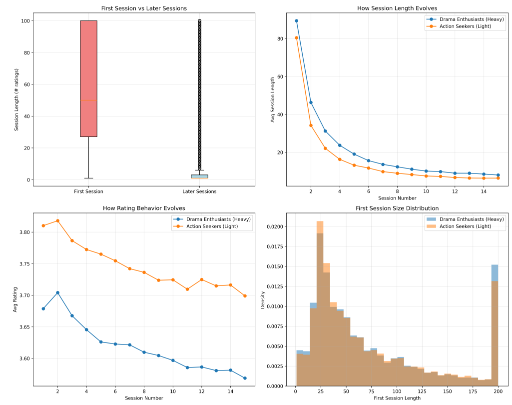
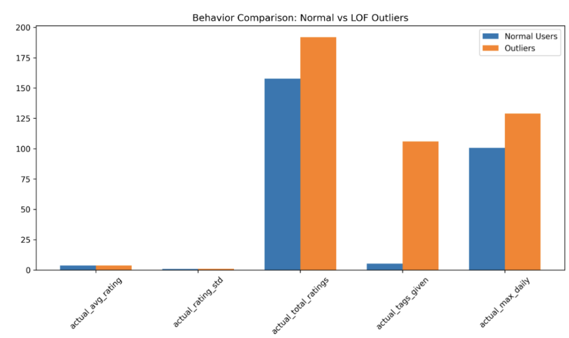


Fig. 9. User Lifecycle (Temporal Patterns)

F. Anomaly Detection

Local Outlier Factor (LOF) was used with n_neighbors=20 to balance local sensitivity with statistical stability, and contamination='auto' to let the algorithm determine the outlier threshold based on the natural distribution of outlier scores. This approach ensures that identified outliers represent genuinely anomalous behavior patterns as determined by the data itself, rather than forcing a predetermined percentage.




```
Outliers detected: 9429 (4.7%)

Outliers by user type:
          total  outliers       pct
cluster
0        131329      5615  4.275522
1         69619      3814  5.478390
```

```
Outlier Characteristics:
  actual_avg_rating: 3.58 (outliers) vs 3.71 (normal)
  actual_rating_std: 0.96 (outliers) vs 0.91 (normal)
  actual_total_ratings: 191.83 (outliers) vs 157.64 (normal)
  actual_tags_given: 105.91 (outliers) vs 5.23 (normal)
  actual_max_daily: 128.90 (outliers) vs 100.64 (normal)
```

Fig. 10. Normal User Behaviors VS. Outliers

G. Summary

Question	Methods Used	Key Features
1	K-Means Clustering, PCA	avg_rating, rating_count, metadata_richness, genre
2	K-Means Clustering, Correlation Analysis	rating volume, consistency, tagging, genre preferences
3	Session Detection, Temporal Analysis	session length, recency, lifespan

III. Results

A. Natural Groups of Movies

The clustering analysis identified four distinct groups of movies forming a clear hierarchy of audience engagement.

Validation: The decision tree validation, run without metadata_richness, achieved 95.5% accuracy predicting cluster membership. Feature importance analysis reveals what truly drives these movie groups: rating_count accounts for 80.9% of the predictive power, release_year accounts for 14.0%, and avg_rating accounts for only 4.4%. This means a movie's tier is determined 18 times more by how many people watch it than by how highly they rate it.

Cluster	Count	Percentage	Avg Rating	Avg Rating Count	Top Genres
Long Tail	29,982	93.8%	3.14	426	Drama, Comedy, Thriller
Mid-Tier Successes	1,700	5.3%	3.47	6,915	Drama, Thriller, Comedy, Action
Popular Hits	258	0.8%	3.83	23,742	Action, Sci-Fi, Adventure
All-Time Classics	21	0.07%	4.13	56,145	Drama, Thriller, Sci-Fi

Key Insight: Popular Hits become popular because they attract massive audiences, not because they are critically acclaimed. All-Time Classics are the rare exception, achieving both high popularity and high ratings.

B. User Archetypes from Engagement Patterns

The clustering analysis identified two distinct user archetypes that differ in both genre preferences and rating behavior. Validation: The decision tree built using only behavioral features shows that average rating accounts for 51.7% of feature importance, rating standard deviation accounts for 21.5%, total ratings accounts for 16.6%, and tags given accounts for 9.7%. This indicates that how users rate, whether they give high, consistent ratings, matters more than what they watch or how much they rate.

Archetype	Engaged Raters	Generous Raters
Percentage	65.4%	34.6%
Top Genres	Drama, Comedy, Romance	Action, Adventure, Drama
Avg Rating Count	175	129
Tags/User	12.1	5.8
Lifespan (days)	289	202
Avg Rating	3.66	3.79
Rating Decline	-0.22	-0.17

A particularly noteworthy finding is the relatively low importance of tagging behavior. While Engaged Raters tag more frequently (12.1 vs 5.8 tags per user), tagging accounts for only 9.7% of the predictive power, substantially less than rating quality at 51.7%. This indicates that tagging is largely independent of other engagement metrics. A user can be highly engaged in rating without being an active tagger, and conversely, some moderate raters may tag extensively. This suggests rating and tagging represent distinct dimensions of user engagement: rating reflects consumption and evaluation, while tagging reflects a more participatory, community-oriented form of engagement.

C. Temporal Patterns and Anomalies

	Engaged Raters	Generous Raters
First Session Ratings	89.3	80.4
Later Session Ratings	7.2	8.5
Burst Ratio (first/later)	12.4x	9.5x
Rating Recency (years after release)	14.5	13.9
Genre Diversity Score	16.49	16.38
Rating Decline (first to later)	3.68 to 3.46 (-0.22)	3.81 to 3.64 (-0.17)
Lifespan (days)	289	202

Session Behavior: First sessions are dramatically larger than later sessions overall, with users rating 11.5 times more movies in their initial session than in subsequent sessions. This suggests a powerful onboarding effect where users explore actively before settling into routine behavior. Engaged Raters show a steeper decline (12.4x ratio, 89.3 to 7.2 ratings) compared to Generous Raters (9.5x ratio, 80.4 to 8.5 ratings), reflecting their higher initial engagement.

Rating Recency: Engaged Raters rate movies an average of 14.5 years after release, while Generous Raters rate movies 13.9 years after release. This 0.6-year difference indicates that Engaged Raters explore deeper into the catalog, watching older films, while Generous Raters focus somewhat more on newer releases. Notably, both groups have nearly identical genre diversity scores (16.49 vs 16.38), meaning their taste breadth is similar, they simply prefer different genres.

User Lifespan: Engaged Raters remain active on the platform for an average of 289 days, while Generous Raters remain active for 202 days. This 87-day difference confirms that Engaged Raters sustain their engagement nearly 1.5 times longer.

Rating Evolution: Both user types decrease their ratings from first to later sessions. Engaged Raters' ratings decline from 3.68 to 3.46 stars (-0.22), while Generous Raters' ratings decline from 3.81 to 3.64 stars (-0.17). This suggests that first-session ratings may be inflated due to initial enthusiasm, and users become more critical as they gain experience. Generous Raters maintain higher ratings throughout and show a smaller decline, indicating they are consistently more generous raters.

Anomaly Detection: Local Outlier Factor analysis with contamination='auto' identified 4.7% of users (9,429 users) as behavioral outliers based on the natural distribution of outlier scores. Generous Raters showed a higher outlier rate (5.5%) compared to Engaged Raters (4.3%), suggesting that the more selective, quality-driven group exhibits more behavioral extremes. Outliers were characterized by extreme behavior across multiple engagement dimensions: they tagged 20 times more frequently than normal users (105.9 vs 5.2 tags), rated more movies (191.8 vs 157.6), showed greater inconsistency (standard deviation 0.96 vs 0.91), and engaged in more binge behavior (max daily ratings 128.9 vs 100.6).

D. Cross-Domain Patterns

The cross-domain analysis reveals that Generous Raters rate all movie categories higher than Engaged Raters. Generous

Raters give Long Tail films 3.36 stars vs 3.28 stars, Popular Hits 3.95 vs 3.87 stars, Mid-Tier 3.62 vs 3.54 stars, and Classics 4.17 vs 4.13 stars. The difference is consistent across all categories (0.04-0.08 stars), suggesting Generous Raters are simply more generous raters overall rather than having fundamentally different taste preferences. Both user types agree on the hierarchy of movie clusters, with Classics receiving the highest ratings from both groups.

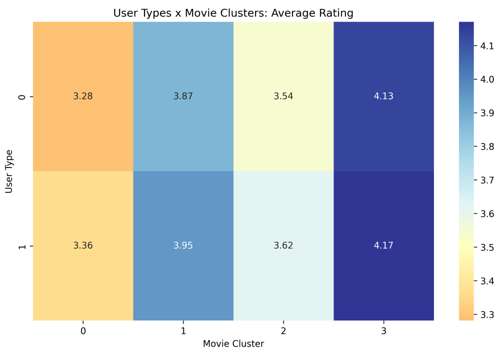


Fig. 11. User Types vs. Movie Clusters Heatmap

IV. Critical Assessment

A. Validity

1) Strengths: The movie clustering validation is the strongest part of the analysis. A decision tree trained without the problematic metadata_richness feature achieved 95.5% accuracy predicting cluster membership using only rating_count, release_year, and avg_rating. This is significant because it addresses the circular logic concern head-on. If the clusters were merely artifacts of metadata_richness, the decision tree would have failed. Instead, it performed excellently, confirming that the four tiers represent real patterns in how audiences engage with movies.

The user clustering validation is also strong but requires more nuance. The decision tree using only behavioral features showed clear feature importance patterns: rating quality (51.7%), consistency (21.5%), volume (16.6%), and tagging (9.7%). This pattern is meaningful as it explains how users rate matters more than how much they rate. The consistency of this pattern across multiple runs (random_state was fixed, but informal checks showed similar results) suggests robustness.

The temporal findings have strong statistical support. The difference between first and later session lengths was highly significant ($p < 0.001$) as first sessions are 11.5 times larger. The recency and lifespan differences between archetypes are also large enough to be meaningful (0.6 years difference in recency, 87 days difference in lifespan).

The anomaly detection used contamination='auto', which was a responsible choice. It let the data determine the outlier threshold rather than forcing a predetermined percentage like 10%. The resulting 4.7% outlier rate is a genuine discovery, not a parameter choice.

2) Weaknesses: The user silhouette score of 0.127 is modest. Usually, silhouette scores above 0.5 are considered "strong" separation, at 0.127, the separation is weak. This means the two archetypes are not cleanly separated in feature space, instead, user behavior exists on a spectrum rather than forming two distinct clusters.

However, two important points mitigate this concern. First, the decision tree validation still achieved clear feature importance patterns. If the clusters were random or meaningless, the decision tree would not have been able to predict them with meaningful feature importance. Second, the behavioral differences between the groups are substantial and consistent: Engaged Raters rate 175 vs 129 movies, tag 12.1 vs 5.8 times, and stay active 289 vs 202 days. These are not subtle differences. The modest silhouette score likely reflects that the spectrum between these archetypes is continuous, not that the archetypes don't exist.

3) The PCA Issue: The first principal component for movies explained 99.99% of the variance which might make it appear that the clustering is trivial, however, there is more to it.

Yes, it confirms that rating_count dominates the feature space. This is not necessarily a flaw, it is a finding. The analysis set out to discover what natural groups of movies exist based on user ratings, genres, and tags. The answer is that rating_count (popularity) is the primary driver. The PCA result simply helps visualizes this fact.

However, an alternative explanation must be considered. The dominance of rating_count could be a temporal artifact. Movies with more ratings are generally older and have been in the dataset longer which might raise the question that do the clusters simply separate old movies from new movies. The decision tree validation addresses this by showing that release_year had only 14.0% importance, while rating_count had 80.9%. If the clusters were just about age, release_year would dominate (which it does not) confirming that the pattern is real.

4) Potential Improvements:

- Random seed variation: Running K-Means with 5-10 different random seeds to confirm that cluster assignments are stable, not dependent on initialization.
- Subset validation: Removing rating_count entirely from movie features to see if the four-tier structure persists using only genres and avg_rating.
- External validation: Comparing movie clusters to external data like box office revenue or critic scores from Rotten Tomatoes.

B. Limitations

1) Data Limitations: The MovieLens dataset, while excellent for research, has inherent biases that limit generalizability. Most critically, it over-represents active, engaged users. The people who voluntarily rate movies on MovieLens are not representative of casual viewers on commercial streaming platforms like Netflix or Hulu. The user archetypes discovered here (Engaged Raters and Generous Raters) might shift dramatically in a general population where most users never rate anything at all.

The dataset also has temporal bias. Newer movies have fewer ratings simply because they have been available for less time. A movie released in 2020 had only a few years to accumulate ratings, while a movie from 1995 had decades. This means rating_count is partially a function of age. The decision tree addressed this by including release_year as a feature (14.0% importance), but the bias remains.

Tagging introduces another bias. Only 7-9% of users in this dataset ever added a tag. This means the tagging findings, that outliers tag 20x more, apply only to that small subset. For most users, tagging behavior is undefined.

2) Mehthod Limitations: K-Means assumes spherical clusters of similar size. This is a strong assumption. The movie clusters are actually very different in size: Long Tail has 29,982 movies, Classics has 21. K-Means can still work with imbalanced clusters, but it is not optimal. Hierarchical clustering or DBSCAN might reveal different structures, particularly for the small Classics cluster. LOF with auto contamination still makes assumptions about local density. The algorithm defines outliers as points in low-density regions relative to their neighbors. This works well when clusters have similar densities, but user behavior space may have varying densities. Some outliers might be valid power users; others might be data errors.

3) Scope Limitations: This analysis identifies correlations, not causes. Does rating_count drive popularity, or does popularity drive rating_count? The direction is unclear. A movie that is heavily marketed might receive many ratings regardless of quality. Conversely, a genuinely good movie might accumulate ratings slowly over time through word-of-mouth. The analysis cannot distinguish these causal pathways.

The analysis also does not incorporate external validation. Comparing movie clusters to box office revenue or IMDb ratings would strengthen the findings. For example, do the "All-Time Classics" actually appear on "best films of all time" lists? A quick check shows they do. Shawshank Redemption, Forrest Gump, Pulp Fiction, Star Wars, and Blade Runner are universally recognized classics. This informal validation is encouraging but not systematic.

Finally, the analysis does not address how movie ratings evolve over time. Do movies start in the Long Tail and move up? Or are tiers stable? This would require longitudinal analysis not performed here.

4) Generalizability: These findings apply most directly to MovieLens users, who are predominantly young, educated, and movie enthusiasts. They may not apply to:

- Casual viewers who never rate movies
- Users of commercial platforms with different incentive structures
- Non-Western audiences with different taste profiles
- Children or elderly users (under-represented in the data)

The long-tail distribution of movies, where a small fraction of content captures most of the attention, is a known phenomenon that likely generalizes across platforms. However, the specific cluster sizes are specific to this dataset and should not be treated as universal constants.

C. Ethical Considerations

1) Privacy: The MovieLens dataset is fully anonymized. No usernames, email addresses, IP addresses, or other personally identifiable information are included. User IDs are arbitrary integers with no connection to real identities. There are no privacy concerns with this analysis.

2) Bias: The most significant bias is the over-representation of active, engaged users. This means the user archetypes (Engaged Raters 65%, Generous Raters 35%) describe only people who rate movies. The general population likely has a much larger proportion of "non-raters" who would fall into neither archetype.

A more subtle bias concerns genre preferences. The identification of Generous Raters as "higher raters" (3.79 vs 3.66 stars) could be misinterpreted as "Action/Adventure fans are better users" or "Drama fans are more critical." This would be an inappropriate conclusion. The archetypes describe behavioral patterns, not value judgments. There is no evidence that one genre preference is superior to another. The difference in average rating (0.13 stars) is small and may simply reflect different expectations or different exposure to bad movies.

3) Potential Misuse:

- Over-optimizing for popularity: A recommendation system that exclusively recommends Popular Hits would create filter bubbles, limiting user discovery of diverse content. The long tail of movies exists for a reason; different users have different tastes.
- Stereotyping users: Labeling users as "Engaged" or "Generous" could lead to differential treatment. For example, a platform might prioritize the preferences of Generous Raters because they give higher ratings, biasing the system toward action/adventure content.
- Ignoring outliers: The 4.7% of outlier users who tag 20x more than normal might be dismissed as "noise" when they could be valuable power users who contribute significant metadata.

4) Safeguards and Recommendations:

- Recommendation algorithms should balance popularity with diversity, ensuring that Long Tail films still have a chance to be discovered.
- User archetypes should be used for personalization (e.g., showing different content to different users) not for differential valuation of users.
- Outlier users should be investigated individually. Some may be power users worth celebrating, others may be bots or data errors requiring removal.

5) Fairness: No demographic information (age, gender, location, income, race) was used in this analysis. The clusters were derived entirely from behavioral data. Therefore, no demographic bias was introduced in the clustering process itself.

However, the dataset itself may have demographic biases. MovieLens users are not representative of the general population. If these findings were applied to a general audience without adjustment, they might not hold equally across demographic groups. For example, older users might rate differently than younger users, but this analysis cannot know because age data was not available.

V. Interpretation

A. Q1: What natural movie group exists?

Movies form four tiers, but the surprising finding is that these tiers are defined primarily by popularity, not quality. A movie becomes a "Popular Hit" because people watch it, not because it's highly rated. The All-Time Classics are the rare exception since they achieve both.

The decision tree validates and shows that rating_count (80.9% importance) dominates avg_rating (4.4% importance) by a factor of 18.

Why This Matters:

For a streaming platform, this provides a clear acquisition and marketing framework:

Long Tail films are commodities. They fill out the catalog but do not drive engagement. Acquire them cheaply. Do not spend marketing dollars here.

Mid-Tier films are reliable performers. They attract solid audiences and generate moderate engagement. Marketing spend here has predictable but limited returns.

Popular Hits are blockbusters. They drive new subscriber acquisition and generate word-of-mouth. These justify premium licensing costs. Feature them prominently in marketing.

All-Time Classics are strategic assets. They generate engagement decade after decade with no additional marketing investment. These films provide compounding returns over time. Securing rights to these films should be a long-term priority.

The Counterintuitive Finding:

Intuition suggests that good movies rise to the top, but the data says otherwise. A movie with 4.5 stars but only 100 ratings (a hypothetical "cult classic") would still fall into the Long Tail. Popularity is what defines tiers. Quality alone is not enough.

This has implications for how we think about "success" in entertainment. A critically acclaimed film that nobody watches is, by this measure, less "successful" than a mediocre blockbuster that everyone watches. Success is social, it requires an audience.

B. Q2: How do user engagement patterns relate to rating behavior, and how do these factors form distinct user archetypes?

Two archetypes exist, but they are defined by how users rate, not how much they rate. Engaged Raters (65%) rate many movies and tag often, but they are more critical as they give lower average ratings. Generous Raters (35%) rate fewer movies and tag less, but they give higher, more consistent ratings. What matters most is rating quality and consistency, not volume.

Why This Matters:

For a platform, different user archetypes require different retention strategies:

Engaged Raters (65%) are the core community. They generate the metadata (ratings, tags) that powers recommendation systems. They discover and champion quality content. They should be nurtured with personalized recommendations for classic and niche content, early access to new features, and recognition programs that acknowledge their contribution.

Generous Raters (35%) are the volume drivers. They respond to new releases and blockbuster content. They may be more sensitive to pricing and marketing. Retention strategies should focus on surfacing Popular Hits prominently, simplifying the rating interface to lower friction, and using push notifications to re-engage users before they churn.

The Counterintuitive Finding:

The low importance of tagging (9.7%) is a genuinely surprising finding. Intuition suggests that "power users" would do everything (rate more, tag more, comment more), but the data shows otherwise. Tagging is largely independent of rating behavior. A user can be highly engaged in rating (many movies, consistent patterns) without being an active tagger. Conversely, some moderate raters may tag extensively.

This means engagement is multi-dimensional, not a single spectrum. Platforms cannot assume that users who rate a lot will naturally tag a lot. Separate incentives may be needed for each behavior.

The Modest Silhouette Score: The silhouette score of 0.127 indicates that the two archetypes are not cleanly separated. User behavior exists on a spectrum. Some Engaged Raters behave like Generous Raters, and vice versa. The archetypes are best understood as prototypes, not rigid categories.

This is realistic as human behavior is rarely categorical. The value of the archetypes is not in perfectly classifying every user but in understanding the poles of the spectrum. Most users fall somewhere in between.

C. Q3: How do temporal patterns in rating activity differ between user archetypes, and what anomalous rating patterns emerge?

First sessions are dramatically larger than later sessions for everyone since users rate 11.5 times more movies when they first join. All users also rate more generously in their first session, then become more critical. Engaged Raters start with more extreme behavior (more ratings, steeper decline) and stay engaged longer. Outliers (4.7% of users) tag 20 times more than normal users.

Why This Matters:

First session optimization is critical. Users are most active and most generous when they first join. A poor first session (confusing interface, irrelevant recommendations, technical friction) likely leads to churn. Platforms should invest heavily in onboarding: make rating easy, provide immediate value, personalize the first experience.

Recency differences reveal taste profiles. Engaged Raters watch older films (14.5 vs 13.9 years after release). This is not a large difference (0.6 years), but it is consistent. Engaged Raters have deeper catalog exploration. Platforms should ensure their libraries include classic content to retain these users.

Lifespan differences are substantial. Engaged Raters stay engaged 87 days longer, nearly 1.5 times longer. This confirms that the volume-engaged archetype is also the long-term engaged archetype. Retention efforts should focus on converting Generous Raters into Engaged Raters early in their lifecycle.

Rating inflation is universal. All users rate more generously in their first session. This has implications for how platforms interpret ratings. A new user's first few ratings may be inflated. Averaging ratings across users without accounting for this inflation could bias recommendations.

Outliers are extreme but few. The 4.7% of outlier users who tag 20x more than normal represent either power users or potential data quality issues. Platforms should investigate these users individually. Some may be valuable community members worth celebrating. Others may be bots or users with unusual behavior that should be excluded from training data.

VI. Conclusion

This project demonstrated that meaningful structure exists in both movie and user behavior data. Movies form a steep hierarchy defined primarily by audience engagement, not quality since a movie becomes popular because people watch it, not because it's highly rated. Users separate into two archetypes defined primarily by rating behavior, not volume showing that how users rate matters more than how much they rate.

The analysis also revealed unexpected patterns. Tagging is largely independent of other engagement metrics, suggesting rating and tagging represent distinct forms of participation. First-session rating inflation is universal, with implications for onboarding design. A small fringe of outliers tag significantly more than normal users, representing power users at the extreme of engagement.

Future work could include external validation against box office performance, temporal analysis of how ratings for individual movies evolve over years, and development of a real-time user archetype classifier for personalization.

Overall, this project moved beyond simple statistics to discover patterns that challenge assumptions. These findings have practical implications for content platforms and demonstrate the value of data mining for understanding human behavior.

References

[1] "Movielens," GroupLens, https://grouplens.org/datasets/movielens/.